Daniel Coblentz

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EDUCATION

Hood College — Frederick, MD — Expected 2026
Bachelors of Science, Computer Science | Minor: Mathematics — GPA: 3.8/4.0 (Major GPA: 3.96)

Honors & Awards: Trailblazer Scholar (merit scholarship), Dean's List (all semesters), Volpe Scholar (top 2%), Christine P. Tischer Scholar (top 2%), Upsilon Pi Epsilon Honor Society (top 5%).
Related Coursework: Data Structures & Algorithms, Object-Oriented Programming I/II, Database Management Systems, Artificial Intelligence, Statistical concepts, Calculus I & II, Linear Algebra, Probability, Discrete Math.

Frederick Community College — Frederick, MD — 2021 - 2023
Associates of Arts, General Studies — GPA: 3.5/4.0

Related Coursework: Introduction to Programming, College Algebra, Statistics, Precalculus

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, SQL (PostgreSQL, MySQL), JavaScript, R, Bash,
Frameworks & Tools: Spring Boot, React, Node.js, Docker, Git, GitHub Actions, Postman, CI/CD, API Design
Cloud & Deployment: AWS (EC2, ECS, Lambda, CodeDeploy), Google Cloud Platform, Heroku, Vercel

PROFESSIONAL EXPERIENCE

Machine Learning Research Intern - Lawrence Berkeley National Laboratory — **June 2025 – Present**

- Built a multimodal machine learning pipeline that combines patient medical records and clinical notes to improve risk prediction for sleep apnea patients.
- Designed and evaluated data fusion methods using TabNet and transformer-based text encoders, achieving 88% AUC in phenotype prediction.
- Ran controlled experiments with Logistic Regression, Random Forest, and Kolmogorov–Arnold Networks, optimizing performance through hyperparameter tuning and model evaluation.
- Processed and analyzed the MIMIC-IV dataset (40K+ patients) using reproducible data pipelines and SHAP-based feature analysis to improve interpretability.

Undergraduate Teaching Assistant - Hood College CS Department — **August 2024 - May 2025**

- Instructed over 70+ students in Data Structures & Algorithms, Discrete Mathematics and advanced programming fundamentals covering topics such as hashing, graphs, trees, sorting, Queues and OOP.
- Conducted code reviews on projects and assignments, and provided detailed feedback including best practices for back-end development to help students improve their coding skills.
- Developed custom problem sets and held weekly review sessions to reinforce core CS concepts.

Analyst Intern – International HELP — **May 2024 – August 2024**

- Analyzed 10K+ health survey responses with SQL and Pandas to uncover key factors driving engagement.
- Built KPI metrics and trend models that identified 3 high-risk regions and guided outreach strategies.
- Delivered reports and dashboards to senior leaders, enabling faster and smarter data-driven decisions.

LEADERSHIP & ACTIVITIES

Competitive Programming Club – Co-Lead — **October 2024 - Present**

- Served as Co-lead managing weekly meetings and developing strategies in problem solving advanced algorithms.
- Competed in timed programming contests and secured Top 3 placements at CCSC-Eastern Regionals.
- Mentored new members in C++ and Python techniques for competitive programming.

Computer Science Club – Active Member — **November 2024 – Present**

- Assisted in organizing weekly technical workshops and peer-learning sessions for CS students.
- Recruited 10 new members by promoting events and sharing academic and career resources on campus.
- Supported hands-on coding sessions to help students strengthen programming and problem-solving skills.